

AI - Assignment 3

Chess AI using Minimax and Alpha-Beta Pruning

Team members: CS22B011 - B. Satyanarayana
CS22B019-D. A . Raghuv eer

Implemented Algorithms:

- **Minimax Algorithm:**
Explores every possible move to a specified depth assuming optimal play by both sides.
- **Alpha-Beta Pruning:**
Optimized Minimax — prunes subtrees that won't affect the final decision, reducing computation



Evaluation Function:

- Assigns scores based on piece values:
Pawn=100, Knight=320, Bishop=330, Rook=500, Queen=900, King=20000
- Additional Conditions:
 - +9999 for checkmate,
 - 0 for stalemate/insufficient material

Observation:

- Both algorithms produced the **same gameplay** due to:
 - Identical evaluation logic
 - Shallow depth (depth = 2)

Differences and results observed:

<u>Criteria</u>	<u>minimax</u>	<u>alpha-beta</u>
Gameplay	Identical	Identical
Speed	Slower	Faster
Pruning used(?)	NO	YES
Outcome Quality	Same	Same

Conclusion:

- Alpha-Beta is **computationally better**, but same outcome due to shared logic.
- Evaluation function solely based on **material value**, no positional understanding